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CENTRAL ADMINISTRATIVE FLEET MANAGEMENT SYSTEM FOR UPDATING IN-FLIGHT ENTERTAINMENT AND COMMUNICATIONS SOFTWARE

Abstract

Systems, devices, and methods updating in-flight entertainment and communications (IFEC) systems in aircraft are disclosed herein. In some implementations, a central administrative fleet management system includes a database configured to store an aircraft installation plan and a managing processor operably coupled to the database. The aircraft installation plan can define a planned set of software content to be stored on an IFEC system of a particular aircraft. The managing processor can be configured to receive an identity of the aircraft, identify a portion of the aircraft installation plan corresponding to the aircraft based on the received identity of the aircraft, and communicate the identified portion of the aircraft installation plan to the IFEC system. The IFEC system can then be updated using software content received from a portable content loader operably coupled to the IFEC system based on the identified portion of the aircraft installation plan.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure is directed generally to systems, methods, and devices for updating in-flight entertainment and communications software.

BACKGROUND

[0002] Air travel typically involves journeys over extended distances that at the very least take several hours to complete. Airlines thus often accommodate its customers with in-flight entertainment and communications (IFEC) systems that provide movies, TV shows, music, games, flight tracking, and other programs for passengers to interact with during the flight. Ideally, IFEC systems are updated frequently such that airplanes can always carry the latest and most popular multimedia content. However, updating IFEC systems conventionally requires maintenance personnel or others to physically carry a storage device containing such multimedia content onto the airplane, and to manually transfer select content to the airplane. This can be a labor-intensive and time-consuming process that prohibits frequent updating. Accordingly, there is a need in the art for updating aircraft IFEC systems with software updates, recent media content, etc. in a faster, more efficient, and more reliable manner.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Features, aspects, and advantages of the presently disclosed technology may be better understood with regard to the following drawings.

[0004] FIG. 1 shows an example of an in-flight entertainment and communications (IFEC) system installed in an airplane based on some implementations of the disclosed technology.

[0005] FIG. 2 shows an example block diagram of a computing device based on some implementations of the disclosed technology.

[0006] FIG. 3 is a schematic diagram of an environment in which an IFEC system can be updated based on some implementations of the disclosed technology.

[0007] FIG. 4 is a schematic diagram of an example process of updating an IFEC system based on some implementations of the disclosed technology.

[0008] FIG. 5 is a flowchart illustrating a method for updating IFEC software based on some implementations of the disclosed technology.

[0009] A person skilled in the relevant art will understand that the features shown in the drawings are for purposes of illustrations, and variations, including different and/or additional features and arrangements thereof, are possible.

DETAILED DESCRIPTION

[0010] Various implementations of the disclosed technology provide techniques for automating the process of updating software content in in-flight entertainment and communications (IFEC) systems and devices on airplanes. Air travel typically involves journeys over extended distances that at the very least take several hours to complete. Some of the longer non-stop international flights have scheduled durations of over sixteen hours with travel distances extending beyond ten thousand miles. Passengers on board the aircraft are confined within an enclosed space of a designated seat for the entire duration of the flight, with only a few limited opportunities to leave the seat for use of the lavatory and so forth. Thus, even on the shortest trips an airline passenger has some idle time, which the passenger may occupy with work, leisure, and/or rest.

[0011] Many passengers bring their own personal electronic devices such as smart phones, media players, electronic readers, tablets, laptop computers, and so forth, for the express purpose of

keeping occupied, but airlines also accommodate its customers with in-flight entertainment and communications (IFEC) systems. Although the specific installation may vary depending on the service class, each passenger seat is equipped with a display device, an audio output modality, an input modality such as a remote control, and a terminal unit. Generally, the terminal unit may generate video and audio signals, receive inputs from the remote control, and execute pre-programmed instructions in response thereto. The display device is typically an LCD screen that is installed on the seatback of the row in front of the passenger, though in some cases it may be mounted to a bulkhead or retractable arm or the like that is in turn mounted to the passenger's seat. Furthermore, the audio output modality is a headphone jack, to which a headphone, either supplied by the airline or by the passenger, may be connected.

[0012] Via the display and the audio outputs, a wide variety of multimedia content can be presented to the passenger. Recently released movies are a popular viewing choice, as are television shows such as news programs, situation and stand-up comedies, documentaries, and so on. Useful information about the destination such as airport disembarking procedures, immigration and custom procedures and the like is also frequently presented. Audio-only programming is also available, typically comprised of playlists of songs fitting into a common theme or genre. Likewise, video-only content such as flight progress mapping, flight status displays, and so forth are available. Many in-flight entertainment systems also include video games that may be played by the passenger using the remote control, which may also have alternative uses, namely, for navigating through the vast multimedia content library and making selections thereof for viewing and/or listening. Thus, the terminal unit may also include a content selection application with a graphical user interface, through which such navigation of the multimedia content library is possible. The foregoing types of programming that can be presented to the passenger via the in-flight entertainment system will henceforth be generally referred to as multimedia content.

[0013] The multimedia content is encoded and stored as digital data, with a video decoder and audio decoder of the terminal unit functioning to generate the aforementioned video and audio signals therefrom. It is desirable to have a wide range of different multimedia content to satisfy the varying tastes of passengers. It is also desirable to have a sufficient volume of multimedia content so that passengers can remain occupied with entertainment for the entire duration of the flight. Accordingly, the multimedia content stored onboard the aircraft can range in the hundreds of gigabytes, if not over a terabyte. The majority of the data comprises the video programming, although the audio and video game content may be significant as well. This data is typically not stored on each individual terminal unit, but rather, in a central content server also onboard the aircraft. In this regard, the terminal unit is understood to incorporate networking modalities such as Ethernet to establish data communications with the central content server. Once a particular selection of multimedia content is requested by the passenger via the content selection application, the terminal unit may retrieve the same from the central content server, decode the data, and present it to the passenger.

[0014] As important as variety and volume may be in regards to the multimedia content, novelty is as important for airlines to keep its passengers engaged with the in-flight entertainment system, particularly for valuable frequent fliers. Thus, the multimedia content stored on the content server must be frequently updated. Due to the large volume of data involved, a portable content loader that is generally comprised of a hard disk drive, an optical drive, or a solid state drive loaded with the update data is physically carried onboard while the aircraft is on the ground and connected to the central content server. A download or copy process is then initiated, and once complete, the portable content loader is disconnected and removed from the aircraft.

[0015] Different aircraft can have different software content requirements and permissions. For example, geographic restrictions, airline licensing agreements, cultural differences, language preferences, the aircraft model, the anticipated in-flight connectivity, the flight schedule, or passenger data associated with a particular aircraft and its flight route may affect what software

content, such as flight tracking program, movies, music, TV shows, or the like, should or can be installed. In conventional settings, personnel such as a maintenance technician would reference paperwork and manually confirm what software content is currently on the airplane, what software content needs to be added or removed, in what order any software updates must be made, etc. This can be a labor intensive process that takes a long time and involves significant human decision-making, making the process prone to human errors.

[0016] Also in part because of the laborious manual procedures involved, this update process typically takes place on a periodic schedule, preferably during a layover between flights, such as when aircraft maintenance is conducted. It would be desirable for new multimedia content to be made available on a more frequent basis, incorporating programming that may be only days or even a few hours old. Yet, the expense and labor involved with the use of specialized content loader devices may preclude such an aggressive update schedule, and these issues are particularly acute for large fleets of aircraft. Also in part because of the laborious manual procedures involved, this update process can be vulnerable to human errors.

[0017] Accordingly, there is a need in the art for updating aircraft IFEC systems with software updates, recent media content, etc. in a faster, more efficient, and more reliable manner. There is also a desire to shift some of the decision-making from personnel to an automated system.

[0018] In some implementations of the present technology, a central administrative fleet management system automates much, if not all, of the decision-making involved in the IFEC software content update process. The fleet management system can be ground-based and can maintain an aircraft installation plan that defines what airplanes should be storing what software content. Thus, when a particular airplane communicates with the fleet management system, the fleet management system can identify the portion of the aircraft installation plan that corresponds to that particular airplane. The fleet management system can then communicate the identified portion of the aircraft installation plan to the IFEC system on the airplane such that the IFEC system can reference the software content already stored in the IFEC system to determine what software content is missing and what software content needs to be added. by automating the software content update process, the fleet management system can significantly reduce any human errors and increase the overall reliability and speed of the process.

[0019] Various implementations will be discussed in detail with reference to the figures below. In the description, the technology is described with respect to aircraft such as commercial planes, but the implementations of the disclosed technology can be applicable to other vehicles such as jets, buses, trains, ships, and other types of passenger vehicles.

[0020] FIG. 1 shows an example of an in-flight entertainment and communications (IFEC) system **110** installed in an airplane **102** based on some implementations of the disclosed technology. The IFEC system **110** includes a plurality of in-flight entertainment (IFE) devices **108**, which can provide various entertainment and connectivity services, including video and audio streaming and Internet communications, to passengers on board. For example, the IFE devices **108** can include such as monitors, handsets, ports for transferring audio/power/data, etc. The IFE devices **108** can also communicate or otherwise interact with personal electronic devices (PEDs) carried by passengers. PEDs may refer to any electronic computing device that includes one or more processors or circuitries for implementing the functions related to data storage, video and audio streaming, wired communications, wireless communications, etc. Examples of the PEDs include cellular phones, smart phones, tablet computers, laptop computers, and other portable computing devices. In implementations of the disclosed technology, PEDs may have the capability to execute application software programs (“apps”) to perform various functions and/or communicate with the IFE devices **108**.

[0021] In the illustrated embodiment, the passenger seats are individually labeled Seat **11** to Seat **66**. In some implementations, the IFE devices **108** are provided at each passenger seat, such as located at each of the seatbacks of the passenger seats, and/or on cabin walls and/or deployable

from an armrest for seats located at a bulkhead (i.e., in the first row of a section). The IFE devices **108** can include displays providing interfaces to each passenger through which each passenger enters their selections on the entertainment option, e.g., the particular selections, emergency requests, etc. Upon receiving the selection from the passengers, based on the selections from the passengers, the IFE devices **108** can display entertainment content and travel information.

[0022] A server **122** can be communicably coupled with the IFE devices **108** and/or the PEDs via one or more wireless access points **120**, and perform various operations including updating various software content as discussed with reference to FIGS. **3** and **4**. The communications between the server **122**, the IFE devices **108**, and the PEDs are realized by wired and/or wireless connections. In some implementations, the communication among the server **122**, the IFE devices **108**, and the PEDs are achieved through the antenna **124** to and from a ground-based server **114** and/or ground-based cell towers by, for example, a provision of network plugs at the seat for the IFE devices **108** to a wired onboard local area network. In some other implementations, the communications among the server **122**, the IFE devices **108**, and the PEDs are achieved through the antenna **126** to and from satellites **130**, **132**, **134** in an orbit (e.g., via a cellular network utilizing one or more onboard base station(s), Wi-Fi utilizing the wireless access point **120**, and/or Bluetooth).

[0023] The server **122**, the IFE devices **108**, and the PEDs form a local network on board the airplane **102** through an onboard router (not shown). The server **122** is also communicably coupled with the ground server **114** through the antenna **124** for receiving and transmitting information from/to the ground server **114**. The ground server **114** can be located at various locations, such as a computer center at an arbitrary location on the ground, etc. The ground server **114** may be in communication with a database **116**, provide information from the database **116** to the server **122**, and store information received from the server **122** in the database **116**. Although FIG. **1** shows that the database **116** is provided separately from the ground server **114**, the database **116** can be provided as a part of the ground server **114**.

[0024] FIG. **2** shows an example block diagram of a computing device **200** (e.g., an onboard server, a ground server, or a portable server) based on some implementations of the disclosed technology. The computing device **200** includes at least one processor **201**, a memory **203**, a transceiver **210**, a control module **220**, a database **230**, and an input/output (I/O) interface **240**. In other embodiments, additional, fewer, and/or different elements may be used to configure the computing device **200**. The memory **203** may store instructions and applications to be executed by the processor **201**. The memory **203** is an electronic holding place or storage for information or instructions so that the information or instructions can be accessed by the processor **201**. The memory **203** can include, but is not limited to, any type of random access memory (RAM), any type of read-only memory (ROM), any type of flash memory, such as magnetic storage devices (e.g., hard disk, floppy disk, magnetic strips, etc.), optical disks (e.g., compact disc (CD), digital versatile discs (DVD), etc.), smart cards, flash memory devices, etc. The instructions upon execution by the processor **201** configure the computing device **200** to perform the operations (e.g., the operations as shown in and described with reference to FIGS. **3-5**), which will be described in this patent document. The instructions executed by the processor **201** may be carried out by a special purpose computer, logic circuits, or hardware circuits. The processor **201** may be implemented in hardware, firmware, software, or any combination thereof. The term “execution” is, for example, the process of running an application or the carrying out of the operation called for by an instruction. The instructions may be written using one or more programming language, scripting language, assembly language, etc. By executing the instruction, the processor **201** can perform the operations called for by that instruction.

[0025] The processor **201** operably couples with the memory **203**, the transceiver **210**, the control module **220**, the database **230**, and the I/O interface **240**, to receive, send, and process information and to control the operations of the computing device **200**. The processor **201** may retrieve a set of instructions from a permanent memory device, such as a ROM device, and copy the instructions in

an executable form to a temporary memory device that is generally some form of RAM. In some implementations, the computing device **200** can include a plurality of processors that use the same or a different processing technology. The transceiver **210** may include a transmitter and a receiver. In some embodiments, the device **200** comprises a transmitter and a receiver that are separate from one another but functionally form a transceiver. The transceiver **210** transmits or sends information or data to another device (e.g., the IFE devices **108**, a reader device, etc.) and receives information or data transmitted or sent by another device (e.g., another server, a PED, etc.).

[0026] The control module **220** of the computing device **200** is configured to perform operations to assist the computing device **200**. In some implementations, the control module **220** can be configured as a part of the processor **201**. When the computing device **200** communicates with the IFE devices **108** in FIG. 1, the control module **220** can be included in the airplane **102**. In some implementations, the control module **220** can operate machine learning/artificial intelligence (AI) applications that perform various types of data analysis to automate analytical model building. Using algorithms that iteratively learn from data, machine learning applications can enable computers to learn without being explicitly programmed. The machine learning/AI module may be configured to use data learning algorithms to build models to interpret various data received from the various devices or components to detect, classify, and/or predict future outcomes. Such data learning algorithms may be associated with rule learning, artificial neural networks, inductive logic programming, and/or clustering. In some implementations, the control module **220** may assist the computing device **200** to perceive their environment and take actions that maximize the effectiveness of the operations performed by the computing device **200**.

[0027] The I/O interfaces **240** enable data to be provided to the computing device **200** as input and enable the computing device **200** to provide data as output. In some embodiments, the I/O interfaces **240** may enable user input to be obtained and received by the computing device **200** (e.g., via a touch-screen display, buttons, or switches) and may enable the computing device **200** to display information. In some embodiments, devices, including touch screen displays, buttons, controllers, audio speakers, or others, are connected to the computing device **200** via I/O interfaces **240**.

[0028] In some implementations, the computing device **200** comprises a computer-readable medium (e.g., the memory **203**) including processor instructions that, when executed by one or more processors (e.g., the processor **201**), cause the one or more processors to perform an IFEC software content update process as described in further detail below with reference to FIGS. 3-5.

[0029] FIG. 3 is a schematic diagram of an environment **300** in which the IFEC system **110** can be updated based on some implementations of the disclosed technology. The environment **300** can include the airplane **102**, the IFEC system **110** installed on the airplane **102**, and a central administrative fleet management system **320** (“the system **320**”) offboard the airplane **102**. In some implementations, the system **320** is implemented at the ground server **114** and/or the database **116**. The environment **300** can also include a manager personnel **302** who can interact with the system **320**, and an operations personnel **304** who can interact with the IFEC system **110**. As discussed in further detail herein, the system **320** can automate much, if not all, of the decision-making conventionally made by the operations personnel **304**.

[0030] The IFEC system **110** can include a user interface, an installer **314** operably coupled to the user interface **312**, and an IFEC database **316** operably coupled to the installer **314**. The user interface **312** can be a graphical user interface (GUI) and/or an audio-based user interface configured to provide information to and receive user inputs from the operations personnel **304**. The installer **314** can include a processor or other logic component and/or a communications module. The installer **314** can be configured to exchange data with the user interface **312**, the IFEC database **316**, and other devices or systems external to the IFEC system **110**, as discussed in further detail below. The IFEC database **316** can be configured to store various software content, such as flight-tracking programs and multimedia content (e.g., news, movies, games, etc.). In some

implementations, the installer **314** and/or the IFEC database **316** are included in the server **122** illustrated in FIG. **1**. In some implementations, the installer **314** and/or the IFEC database **316** are physically or operably separate from the server **122**.

[0031] The operations personnel **304** can be a maintenance technician or other personnel who physically carries a portable content loader **306** onboard the aircraft **102** and operably couples or otherwise connects the portable content loader **306** to the installer **314** of the IFEC system **110**. The portable content loader **306** can include a hard disk drive, an optical drive, or a solid state drive configured to store most, if not all, of the software content, including any software updates, available to the airline or other entity managing the software content to be stored on the IFEC database **316**. In some implementations, the connection between the portable content loader **306** and the installer **314** is wired such that any data transfer therebetween remains fast and reliable.

[0032] The system **320** can include a manager **324** (also referred to as “the managing processor **324**”) and a database **322** operably coupled to the manager **324**. The manager **324** can include a processor or other logic component and/or a communications module. The database **322** can be configured to store an aircraft installation plan, which can define a planned set of software content to be stored on the IFEC system **110** of each of a plurality of aircraft **102**. Different aircraft can have different software content requirements and permissions, and the aircraft installation plan can list those requirements and permissions for different aircraft. For example, the flight route of a particular aircraft may affect what software content (e.g., flight tracking program, movies, music, TV shows) should be installed (e.g., due to geographic restrictions, airline licensing agreements, cultural differences, language preferences, etc.). In another example, the model of a particular aircraft may affect what software content should be installed (e.g., due to different models of IFE devices **108** installed on the airplane **102**, as they may have monitors of varying dimensions that require movies to be formatted differently). In yet another example, the anticipated in-flight connectivity and/or the flight schedule may affect what software content should be installed (e.g., a minimal anticipated in-flight connectivity may require more multimedia content to be uploaded to the IFEC system **110**, whereas a tight flight schedule may prohibit large data transfers that may take long periods of time). In yet another example, data associated with passengers who will be on the airplane **102** may affect what software content should be installed (e.g., certain passengers may be known to have certain movie preferences). Therefore, the aircraft installation plan can be based on at least one of airline licensing agreements, geographic restrictions, language preferences, anticipated in-flight connectivity, aircraft model, passenger data, or flight schedule. In some implementations, the aircraft installation plan can additionally define a proper sequence in which any software content or updates should be uploaded to the IFEC system **110**.

[0033] In some implementations, the manager **324** can automatically generate and routinely update the aircraft installation plan stored on the database **322**. For example, the manager **324** can keep track of the airline's licensing agreements, concurrent geographic restrictions, passenger information, etc. to generate an up-to-date aircraft installation plan. In some implementations, the manager personnel **302** can assist the manager **324** with generating the aircraft installation plan and/or approve the aircraft installation plan generated by the manager **324**. For example, the system **320** can include a user interface (not shown) that the manager personnel **302** can use to view, edit, and/or approve the aircraft installation plan.

[0034] Once the aircraft installation plan has been approved, the manager **324** can communicate with the airplane **102** (e.g., the IFEC system **110**) to receive an identity of the aircraft. The manager **324** can then cross-reference the received identity with the aircraft installation plan, which can include planned sets of software content for multiple airplanes, to identify a portion of the aircraft installation plan that corresponds to the particular airplane **102** and communicate it to the installer **314** of the IFEC system **110**. In some implementations, the communication between the manager **324** and the installer **314** can be wireless, such as via a Wi-Fi network, a cellular communication network, or a satellite connectivity network.

[0035] When the airplane **102** is on the ground, such as during a layover between flights or when aircraft maintenance is conducted, the operations personnel **304** can physically carry the portable content loader **306** onboard the airplane **102**. The operations personnel **304** can then connect the portable content loader **306** to the installer **314** of the IFEC system **110**. The IFEC system **110** can then initiate a software content update process. In some implementations, the installer **314** configures the user interface **312** to prompt the operations personnel **304** to confirm whether the software content update process should begin.

[0036] First, the installer **314** can communicate with (e.g., read) the IFEC database **316** to perform an inventory check to determine the software content currently stored onboard the airplane **102** (also referred to as “the initial set of software content”). The installer **314** can then cross-reference the initial set of software content currently stored in the IFEC database **316** with the planned set of software content defined in the portion of the aircraft installation plan received from the manager **324**. In some implementations, the user interface **312** is configured to inform the operations personnel **304** of any differences between the initial set of software content and the planned set of software content. In other implementations, the installer **314** proceeds without such communication to the operations personnel **304**. If any software content is missing in the database **316**, as defined by the portion of the aircraft installation plan, the installer **314** can transfer the missing software content from the portable content loader **306** to the IFEC database **316**. If any software content that should not be in the IFEC database **316**, as defined by the portion of the aircraft installation plan, is currently in the IFEC database **316**, the installer **314** can delete or remove such software content accordingly. In some implementations, the software content update process can be performed according to the proper sequence defined by the portion of the aircraft installation plan.

[0037] Once the software content update process has been completed, the installer **314** can configure the user interface **312** to inform the operations personnel **304** that the process has been completed. The operations personnel **304** can then disconnect (e.g., unplug) the portable content loader **306** from the IFEC system **110** and carry the portable content loader **306** back offboard the airplane **102**. In some implementations, the IFEC system **110** can communicate the updated set of software content stored in the IFEC database **316** to the manager **324** or other ground server **A**. A more detailed example of a software content update process is illustrated in and described below with reference to FIG. 4.

[0038] In some implementations, instead of storing the aircraft installation plan with the system **320**, portions of the aircraft installation plan can be embedded in the software content data sets. For example, a particular movie can be embedded with a tag that specifies that the movie should not be on airplanes flying from or to certain countries due to licensing requirements, geographic restrictions, etc. In such implementations, the IFEC system **110** cross-reference the identity of the aircraft with those tags to determine whether certain software content should be added to, removed from, or updated onto the IFEC database **316**.

[0039] FIG. 4 is a schematic diagram of an example process of updating the IFEC system **110** based on some implementations of the disclosed technology. As shown, the database **322** of the system **320** can store an example aircraft installation plan **422**. As discussed above with reference to FIG. 3, the aircraft installation plan **422** can be generated based on at least one of airline licensing agreements, geographic restrictions, language preferences, anticipated in-flight connectivity, aircraft model, or flight schedule. In the illustrated embodiment, the example aircraft installation plan **422** defines planned sets of software content for four different aircraft identities. More specifically, the example aircraft installation plan **422** specifies that (i) airplanes with aircraft identity A should have software contents **1**, **2**, and **4**, (ii) airplanes with aircraft identity B should have software contents **1** and **3**, (iii) airplanes with aircraft identity C should have software content **3**, and (iv) airplanes with aircraft identity D should have software contents **2** and **4**.

[0040] The airplane **102** can have an example aircraft identity A **402**. The aircraft identity A **402** can include the aircraft model or type, flight route, passenger data, and/or other details specific to

the airplane **102**. The airplane **102** can communicate the aircraft identity A **402** to the manager **324** of the system **320** via a wireless connection. In some implementations, the aircraft identity A **402** is communicated to the manager **324** by the IFEC system **110**. The manager **324** can then reference the received aircraft identity A **402** to identify a portion of the aircraft installation plan **422** that corresponds to the airplane **102** and communicate that portion to the installer **314** of the IFEC system **110** (e.g., via a wireless connection).

[0041] As schematically shown within the installer **314** in FIG. 4, the portion **424** of the aircraft installation plan **422** defines a planned set of software content for the airplane **102**, namely software contents **1**, **2**, and **4**. The installer **314** can then read the list of software content currently on the IFEC database **316** (e.g., the initial set of software content). For example, in the illustrated embodiment, the IFEC database **316** includes software contents **1**, **3**, and **4**. Therefore, upon cross-referencing the initial set of software content on the IFEC database **316** with the planned set of software content, the installer **314** can determine that the IFEC database **316** needs software content **2** and should not have software **3**. For example, software content **2** can be a movie produced in the current destination country that is expected to be in demand by the upcoming group of passengers but was likely foreign to most of the passengers on the previous flight, and software content **3** can be a movie that is restricted by the current destination country. Therefore, in response to such determinations, the installer **314** can automatically delete or remove software content **3** from the IFEC database **316** and upload software content **2** from the portable content loader **306** to IFEC database **316**. As shown schematically, the portable content loader **306** can include all of software contents **1**, **2**, **3**, and **4** such that the portable content loader **306** is ready to upload any software content needed by the IFEC system **110**. In some implementations, the IFEC system **110** can inform the operations personnel **304** of any or all of the steps taken or to be taken.

[0042] By automating much, if not all, of the decision-making of what software content should be added to, removed from, or updated on the IFEC database **316** for a particular airplane **102**, the central administrative fleet management system **320** can eliminate or at least significantly reduce the risk of human errors that are normally expected when the operations personnel **304** is making those decisions. For example, automating the software content update process in accordance with implementations of the present technology can reduce the risk of redundancy in software content, since the installer **314** cross-references the portion **424** of the aircraft installation plan **422** and performs an inventory check of what software content is already present in the IFEC database **316**. The system **320** can also increase the overall reliability and reduce the overall time for the software content update process compared to conventional settings in which the operations personnel **304** would rely on paperwork to verify inventory and perform data transfers manually.

[0043] FIG. 5 is a flowchart illustrating a method **500** for updating IFEC software based on some implementations of the disclosed technology. While the method **500** is described below with respect to the implementations described in FIGS. 1-4, one of ordinary skill in the art will appreciate that the method **500** can be practiced with other implementations. Moreover, the steps of the method **500** described below can be performed in an order different from the order in which they are presented, the method **500** can omit some of the steps described, and the method **500** can include additional steps.

[0044] At block **502**, the method **500** can include receiving an identity (e.g., the aircraft identity A **402**) of an aircraft (e.g., the airplane **102**). In some implementations, the identity of the aircraft includes at least one of the aircraft type or model, flight route, passenger data, or flight schedule.

[0045] At block **504**, the method **500** can include identifying a portion of an aircraft installation plan (e.g., the portion **424** of the aircraft installation plan **422**) corresponding to the aircraft based on the received identity of the aircraft. The portion of the aircraft installation plan can define a planned set of software content to be stored on an in-flight entertainment and communications (IFEC) system (e.g., the IFEC system **110**) of the aircraft. In some implementations, the identified portion of the aircraft installation plan further defines a planned sequence by which software

content is to be uploaded to the IFEC system.

[0046] At block **506**, the method **500** can include communicating the identified portion of the aircraft installation plan to the IFEC system of the aircraft. In some implementations, the identified portion of the aircraft installation plan is communicated via a wireless connection, such as via Wi-Fi, cellular, or satellite connections. In some implementations, the identified portion of the aircraft installation plan is communicated from a manager of a central administrative fleet management system to an installer of the IFEC system.

[0047] At block **508**, the method **500** can include updating the IFEC system of the aircraft using software content received from a portable content loader (e.g., the portable content loader **306**) operably coupled to the IFEC system based on the identified portion of the aircraft installation plan. In some implementations, updating the IFEC system includes determining that an initial set of software content stored on the IFEC system includes software content not included in the planned set of software content, and in response, removing the software content included in the initial set of software content but not included in the planned set of software content from the IFEC system. In some implementations, updating the IFEC system includes determining that an initial set of software content stored on the IFEC system does not include software content included in the planned set of software content, and in response, uploading the software content included in the planned set of software content but not included in the initial set of software content to the IFEC system from the portable content loader. In some implementations, updating the IFEC system includes uploading software content received from the portable content loader based on the planned sequence.

[0048] In some implementations, the method **500** further includes generating the aircraft installation plan based on at least one of airline licensing agreements, geographic restrictions, language preferences, anticipated in-flight connectivity, aircraft model, or flight schedule. In some implementations, the method **500** further includes receiving a plurality of identities of a plurality of aircraft, identifying a plurality of portions of the aircraft installation plan corresponding to the plurality of aircraft based on the received identities of the plurality of aircraft, communicating the identified plurality of portions of the aircraft installation plan to corresponding ones of a plurality of IFEC systems of the plurality of aircraft, and updating the plurality of IFEC systems using software content received from one or more portable content loaders operably coupled to the plurality of IFEC systems based on the plurality of identified portions of the aircraft installation plan.

[0049] In accordance with the disclosed technology herein, some embodiments may incorporate technical solutions as provided in the following clauses:

[0050] A central administrative fleet management system for updating in-flight entertainment and communications software on an aircraft, the system comprising: [0051] a database configured to store an aircraft installation plan, wherein the aircraft installation plan defines a planned set of software content to be stored on an in-flight entertainment and communications (IFEC) system of the aircraft; and [0052] a managing processor operably coupled to the database, wherein the managing processor is configured to: [0053] receive an identity of the aircraft; [0054] identify a portion of the aircraft installation plan corresponding to the aircraft based on the received identity of the aircraft; and [0055] communicate the identified portion of the aircraft installation plan to the IFEC system of the aircraft, [0056] wherein the IFEC system of the aircraft is configured to be updated using software content received from a portable content loader operably coupled to the IFEC system based on the identified portion of the aircraft installation plan communicated by the managing processor.

[0057] The system of any of the clauses herein, wherein the IFEC system of the aircraft comprises: [0058] an IFEC database configured to store an initial set of software content; and [0059] an installer operably coupled to the database of the IFEC system, wherein the installer is configured to: [0060] receive the identified portion of the aircraft installation plan from the managing

processor; and [0061] update the IFEC database using the software content received from the portable content loader based on the portion of the aircraft installation plan.

[0062] The system of any of the clauses herein, wherein the installer of the IFEC system is configured to update the IFEC database by performing at least one of: [0063] removing from the IFEC database, in response to a determination that the initial set of software content stored on the IFEC database includes software content not included in the planned set of software content, the software content included in the initial set of software content but not included in the planned set of software content, or [0064] uploading to the IFEC database, in response to a determination that the initial set of software content stored on the IFEC database does not include software content included in the planned set of software content, the software content included in the planned set of software content but not included in the initial set of software content from the portable content loader.

[0065] The system of any of the clauses herein, wherein the aircraft installation plan further defines, for each of a plurality of aircraft, a planned sequence by which software content is to be uploaded to the IFEC system, and wherein the installer of the IFEC system is configured to update the IFEC database by uploading software content received from the portable content loader based on the planned sequence.

[0066] The system of any of the clauses herein, wherein the IFEC system of the aircraft further comprises a user interface configured to communicate with personnel onboard the aircraft and carrying the portable content loader, wherein the installer is configured to update the IFEC database in response to an input received by the user interface.

[0067] The system of any of the clauses herein, wherein the managing processor is further configured to routinely update the aircraft installation plan stored on the database.

[0068] The system of any of the clauses herein, wherein the managing processor is configured to receive the identity of the aircraft from the IFEC system of the aircraft.

[0069] The system of any of the clauses herein, wherein the managing processor is configured to communicate the identified portion of the aircraft installation plan to the IFEC system of the aircraft wirelessly via at least one of Wi-Fi, cellular connection, or satellite connection.

[0070] The system of any of the clauses herein, wherein the managing processor is configured to:

[0071] receive a plurality of identities of a plurality of aircraft; [0072] identifying a plurality of portions of the aircraft installation plan corresponding to the plurality of aircraft based on the received identities of the plurality of aircraft; and [0073] communicate the identified plurality of portions of the aircraft installation plan to corresponding ones of a plurality of IFEC systems of the plurality of aircraft.

[0074] The system of any of the clauses herein, wherein the managing processor is further configured to generate the aircraft installation plan.

[0075] A method for updating in-flight entertainment and communications software on an aircraft, the method comprising: [0076] receiving an identity of the aircraft; [0077] identifying a portion of an aircraft installation plan corresponding to the aircraft based on the received identity of the aircraft, wherein the portion of the aircraft installation plan defines a planned set of software content to be stored on an in-flight entertainment and communications (IFEC) system of the aircraft; [0078] communicating the identified portion of the aircraft installation plan to the IFEC system of the aircraft; and [0079] updating the IFEC system of the aircraft using software content received from a portable content loader operably coupled to the IFEC system based on the identified portion of the aircraft installation plan.

[0080] The method of any of the clauses herein, wherein updating the IFEC system comprises:

[0081] determining that an initial set of software content stored on the IFEC system includes software content not included in the planned set of software content; and [0082] removing the software content included in the initial set of software content but not included in the planned set of software content from the IFEC system.

[0083] The method of any of the clauses herein, wherein updating the IFEC system comprises: [0084] determining that an initial set of software content stored on the IFEC system does not include software content included in the planned set of software content; and [0085] uploading the software content included in the planned set of software content but not included in the initial set of software content to the IFEC system from the portable content loader.

[0086] The method of any of the clauses herein, wherein the identified portion of the aircraft installation plan further defines a planned sequence by which software content is to be uploaded to the IFEC system, and wherein updating the IFEC system comprises uploading software content received from the portable content loader based on the planned sequence.

[0087] The method of any of the clauses herein, further comprising: [0088] generating the aircraft installation plan based on at least one of airline licensing agreements, geographic restrictions, language preferences, anticipated in-flight connectivity, aircraft model, or flight schedule.

[0089] The method of any of the clauses herein, further comprising: [0090] receiving a plurality of identities of a plurality of aircraft; [0091] identifying a plurality of portions of the aircraft installation plan corresponding to the plurality of aircraft based on the received identities of the plurality of aircraft; [0092] communicating the identified plurality of portions of the aircraft installation plan to corresponding ones of a plurality of IFEC systems of the plurality of aircraft; and [0093] updating the plurality of IFEC systems using software content received from one or more portable content loaders operably coupled to the plurality of IFEC systems based on the plurality of identified portions of the aircraft installation plan.

[0094] A computer-readable medium including processor instructions that, when executed by one or more processors, cause the one or more processors to: [0095] receive an identity of an aircraft; [0096] identify a portion of an aircraft installation plan corresponding to the aircraft based on the received identity of the aircraft, wherein the portion of the aircraft installation plan defines a planned set of software content to be stored on an in-flight entertainment and communications (IFEC) system of the aircraft; [0097] communicate the identified portion of the aircraft installation plan to the IFEC system of the aircraft; and [0098] update the IFEC system of the aircraft using software content received from a portable content loader operably coupled to the IFEC system based on the identified portion of the aircraft installation plan.

[0099] The computer-readable medium of any of the clauses herein, wherein the processor instructions cause the one or more processors to: [0100] determine that an initial set of software content stored on the IFEC system includes software content not included in the planned set of software content; and [0101] remove the software content included in the initial set of software content but not included in the planned set of software content from the IFEC system.

[0102] The computer-readable medium of any of the clauses herein, wherein the processor instructions cause the one or more processors to: [0103] determine that an initial set of software content stored on the IFEC system does not include software content included in the planned set of software content; and [0104] upload the software content included in the planned set of software content but not included in the initial set of software content to the IFEC system from the portable content loader.

[0105] The computer-readable medium of any of the clauses herein, wherein the identified portion of the aircraft installation plan further defines a planned sequence by which software content is to be uploaded to the IFEC system, and wherein updating the IFEC system comprises uploading software content received from the portable content loader based on the planned sequence.

[0106] Some of the embodiments described herein are described in the general context of methods or processes, which may be implemented in one embodiment by a computer program product, embodied in a computer-readable medium, including computer-executable instructions, such as program code, executed by computers in networked environments. A computer-readable medium may include removable and non-removable storage devices including, but not limited to, Read Only Memory (ROM), Random Access Memory (RAM), compact discs (CDs), digital versatile

discs (DVD), etc. Therefore, the computer-readable media can include a non-transitory storage media. Generally, program modules may include routines, programs, objects, components, data structures, etc. that perform particular tasks or implement particular abstract data types. Computer- or processor-executable instructions, associated data structures, and program modules represent examples of program code for executing steps of the methods disclosed herein. The particular sequence of such executable instructions or associated data structures represents examples of corresponding acts for implementing the functions described in such steps or processes.

[0107] Some of the disclosed embodiments can be implemented as devices or modules using hardware circuits, software, or combinations thereof. For example, a hardware circuit implementation can include discrete analog and/or digital components that are, for example, integrated as part of a printed circuit board. Alternatively, or additionally, the disclosed components or modules can be implemented as an Application Specific Integrated Circuit (ASIC) and/or as a Field Programmable Gate Array (FPGA) device. Some implementations may additionally or alternatively include a digital signal processor (DSP) that is a specialized microprocessor with an architecture optimized for the operational needs of digital signal processing associated with the disclosed functionalities of this application. Similarly, the various components or sub-components within each module may be implemented in software, hardware or firmware. The connectivity between the modules and/or components within the modules may be provided using any one of the connectivity methods and media that is known in the art, including, but not limited to, communications over the Internet, wired, or wireless networks using the appropriate protocols.

[0108] While this document contains many specifics, these should not be construed as limitations on the scope of an invention that is claimed or of what may be claimed, but rather as descriptions of features specific to particular embodiments. Certain features that are described in this document in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or a variation of a sub-combination. Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results.

[0109] Only a few implementations and examples are described, and other implementations, enhancements and variations can be made based on what is described and illustrated in this patent document.

Claims

1. A central administrative fleet management system for updating in-flight entertainment and communications software on an aircraft, the system comprising: a database configured to store an aircraft installation plan, wherein the aircraft installation plan defines a planned set of software content to be stored on an in-flight entertainment and communications (IFEC) system of the aircraft; and a managing processor operably coupled to the database, wherein the managing processor is configured to: receive an identity of the aircraft; identify a portion of the aircraft installation plan corresponding to the aircraft based on the received identity of the aircraft; and communicate the identified portion of the aircraft installation plan to the IFEC system of the aircraft, wherein the IFEC system of the aircraft is configured to be updated using software content received from a portable content loader operably coupled to the IFEC system based on the identified portion of the aircraft installation plan communicated by the managing processor.

2. The system of claim 1, wherein the IFEC system of the aircraft comprises: an IFEC database configured to store an initial set of software content; and an installer operably coupled to the database of the IFEC system, wherein the installer is configured to: receive the identified portion of the aircraft installation plan from the managing processor; and update the IFEC database using the software content received from the portable content loader based on the portion of the aircraft installation plan.
3. The system of claim 2, wherein the installer of the IFEC system is configured to update the IFEC database by performing at least one of: removing from the IFEC database, in response to a determination that the initial set of software content stored on the IFEC database includes software content not included in the planned set of software content, the software content included in the initial set of software content but not included in the planned set of software content, or uploading to the IFEC database, in response to a determination that the initial set of software content stored on the IFEC database does not include software content included in the planned set of software content, the software content included in the planned set of software content but not included in the initial set of software content from the portable content loader.
4. The system of claim 2, wherein the aircraft installation plan further defines, for each of a plurality of aircraft, a planned sequence by which software content is to be uploaded to the IFEC system, and wherein the installer of the IFEC system is configured to update the IFEC database by uploading software content received from the portable content loader based on the planned sequence.
5. The system of claim 2, wherein the IFEC system of the aircraft further comprises a user interface configured to communicate with personnel onboard the aircraft and carrying the portable content loader, wherein the installer is configured to update the IFEC database in response to an input received by the user interface.
6. The system of claim 1, wherein the managing processor is further configured to routinely update the aircraft installation plan stored on the database.
7. The system of claim 1, wherein the managing processor is configured to receive the identity of the aircraft from the IFEC system of the aircraft.
8. The system of claim 1, wherein the managing processor is configured to communicate the identified portion of the aircraft installation plan to the IFEC system of the aircraft wirelessly via at least one of Wi-Fi, cellular connection, or satellite connection.
9. The system of claim 1, wherein the managing processor is configured to: receive a plurality of identities of a plurality of aircraft; identifying a plurality of portions of the aircraft installation plan corresponding to the plurality of aircraft based on the received identities of the plurality of aircraft; and communicate the identified plurality of portions of the aircraft installation plan to corresponding ones of a plurality of IFEC systems of the plurality of aircraft.
10. The system of claim 1, wherein the managing processor is further configured to generate the aircraft installation plan.
11. A method for updating in-flight entertainment and communications software on an aircraft, the method comprising: receiving an identity of the aircraft; identifying a portion of an aircraft installation plan corresponding to the aircraft based on the received identity of the aircraft, wherein the portion of the aircraft installation plan defines a planned set of software content to be stored on an in-flight entertainment and communications (IFEC) system of the aircraft; communicating the identified portion of the aircraft installation plan to the IFEC system of the aircraft; and updating the IFEC system of the aircraft using software content received from a portable content loader operably coupled to the IFEC system based on the identified portion of the aircraft installation plan.
12. The method of claim 11, wherein updating the IFEC system comprises: determining that an initial set of software content stored on the IFEC system includes software content not included in the planned set of software content; and removing the software content included in the initial set of software content but not included in the planned set of software content from the IFEC system.

- 13.** The method of claim 11, wherein updating the IFEC system comprises: determining that an initial set of software content stored on the IFEC system does not include software content included in the planned set of software content; and uploading the software content included in the planned set of software content but not included in the initial set of software content to the IFEC system from the portable content loader.
- 14.** The method of claim 11, wherein the identified portion of the aircraft installation plan further defines a planned sequence by which software content is to be uploaded to the IFEC system, and wherein updating the IFEC system comprises uploading software content received from the portable content loader based on the planned sequence.
- 15.** The method of claim 11, further comprising: generating the aircraft installation plan based on at least one of airline licensing agreements, geographic restrictions, language preferences, anticipated in-flight connectivity, aircraft model, or flight schedule.
- 16.** The method of claim 11, further comprising: receiving a plurality of identities of a plurality of aircraft; identifying a plurality of portions of the aircraft installation plan corresponding to the plurality of aircraft based on the received identities of the plurality of aircraft; communicating the identified plurality of portions of the aircraft installation plan to corresponding ones of a plurality of IFEC systems of the plurality of aircraft; and updating the plurality of IFEC systems using software content received from one or more portable content loaders operably coupled to the plurality of IFEC systems based on the plurality of identified portions of the aircraft installation plan.
- 17.** A computer-readable medium including processor instructions that, when executed by one or more processors, cause the one or more processors to: receive an identity of an aircraft; identify a portion of an aircraft installation plan corresponding to the aircraft based on the received identity of the aircraft, wherein the portion of the aircraft installation plan defines a planned set of software content to be stored on an in-flight entertainment and communications (IFEC) system of the aircraft; communicate the identified portion of the aircraft installation plan to the IFEC system of the aircraft; and update the IFEC system of the aircraft using software content received from a portable content loader operably coupled to the IFEC system based on the identified portion of the aircraft installation plan.
- 18.** The computer-readable medium of claim 17, wherein the processor instructions cause the one or more processors to: determine that an initial set of software content stored on the IFEC system includes software content not included in the planned set of software content; and remove the software content included in the initial set of software content but not included in the planned set of software content from the IFEC system.
- 19.** The computer-readable medium of claim 17, wherein the processor instructions cause the one or more processors to: determine that an initial set of software content stored on the IFEC system does not include software content included in the planned set of software content; and upload the software content included in the planned set of software content but not included in the initial set of software content to the IFEC system from the portable content loader.
- 20.** The computer-readable medium of claim 17, wherein the identified portion of the aircraft installation plan further defines a planned sequence by which software content is to be uploaded to the IFEC system, and wherein updating the IFEC system comprises uploading software content received from the portable content loader based on the planned sequence.
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